# Multi-scale Emergent Reality Theory (MER): A Covariant Action Principle, Scale-Dependent Topology, and the Epistemic Unification of Quantum and Relativistic Regimes

(Revised Edition, with Corrected Field-Equation Derivations)

Martin Ouimet

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**Abstract**

Modern theoretical physics remains fundamentally bifurcated: Quantum Mechanics (QM) formulates reality in terms of discrete, probabilistic state vectors residing in a Hilbert space $\mathcal{H}$, whereas General Relativity (GR) models spacetime as a continuous, deterministic 4D pseudo-Riemannian manifold $(M, g_{\mu\nu})$. Multi-scale Emergent Reality (MER) Theory proposes that physical laws, observed symmetries, and probability distributions are scale-dependent projections of a single, continuous 4D covariant field action $S_{MER}$. The field state consists of a complex scalar order-parameter field $\Phi(x^\mu)$ coupled to an observer-mismatch vector field $\epsilon^\mu(x^\mu)$ through non-linear potential terms whose coefficients are the conjugate golden-ratio eigenvalues ($\phi \approx 1.6180339$, $\psi \approx -0.6180339$).

This edition (v0.7.0) is a corrected revision of v0.6.0. A line-by-line re-derivation of the scalar field equation (Section 5) found that the field equation published in v0.6.0 did not follow from the stated action: a convective term was retained that in fact cancels identically, and the coefficient of the surviving term was mis-stated. The Madelung/quantum-limit proof (Appendix B) has been re-derived to match the corrected field equation, changing its physical conclusion: the coupling no longer perturbs probability conservation, but instead enters the Hamilton–Jacobi (phase) equation as an additive potential term. The experimental falsification table (Section 16) has been reframed to make explicit that its proportionality constants are not yet numerically fixed by the theory. Everything else is unchanged from v0.6.0 and carries the same epistemic status (Postulate / Proposition / Conjecture) as before. See the *Erratum* immediately after this abstract for an itemized account of what changed and why.

## Erratum and Revision Notes (v0.6.0 → v0.7.0)

This section is new in v0.7.0 and documents, without euphemism, what was wrong in v0.6.0 and what has been changed.

(E1) **Proposition 1 (scalar field equation) did not follow from Postulate 3.** Varying the action term-by-term with respect to $\Phi^*$ shows that the term $2\phi\,\epsilon^\mu\nabla_\mu\Phi$ appearing in the v0.6.0 field equation cancels exactly against a term generated by the $\phi$-piece of the potential; it should never have appeared in the equation of motion. The term that *does* survive is proportional to $\phi - \psi = \sqrt{5}$, with a different numerical coefficient than previously stated. The corrected equation is derived explicitly, by two independent methods, in Section 5.

(E2) **This changes the physical content of the scalar equation.** In v0.6.0, the surviving coupling term carried a piece of the same phase as $\nabla_\mu\Phi$, which is why it appeared to perturb the continuity equation in Appendix B. In the corrected equation, the surviving term is real and proportional to $\Phi$ itself (not its derivative), which means it cannot contribute to the continuity equation at all — it can only modify the phase (Hamilton–Jacobi) equation. Appendix B has been re-derived accordingly.

(E3) **A consequence for Postulate 2.** The corrected derivation shows that only the *difference* $\phi - \psi = \sqrt{5}$ enters the scalar equation of motion; the individual values of $\phi$ and $\psi$ do not appear separately. Any two real numbers $a, b$ with $a - b = \sqrt{5}$ would generate the identical equation of motion. This does not invalidate Postulate 2 as a definition, but it does mean the scalar sector alone provides no evidence that the coupling constants must specifically be the golden-ratio roots, only that their difference must equal $\sqrt{5}$. This is noted where relevant and revisited in the Discussion.

(E4) **Proposition 2 (vector field equation) has not been independently re-verified in this revision.** A parallel calculation suggests the same $\sqrt{5}$ structure should appear, by the same mechanism, in the vector equation, but the overall sign depends on metric and variational sign conventions that were not exhaustively checked. It is left in its v0.6.0 form with a flag rather than silently amended (Section 5).

(E5) **Appendix A (stress-energy tensor) has not been independently re-derived in this revision.** It is carried over unchanged from v0.6.0. Given the error found in Proposition 1, the interaction term $T_{\mu\nu}^{(int)}$ should be treated as unverified pending a dedicated re-derivation.

(E6) **Section 16 (falsification table) has been reframed, not newly derived.** None of the four predicted-signal rows had an explicit proportionality constant in v0.6.0; the coupling $\kappa$ carries only a uniform prior on $[0, 1]$ (Section 17.2), so as stated the table could not have actually ruled anything in or out. The table is retained as a statement of *functional form* with an explicit caveat, rather than presented as calibrated numerical thresholds.

(E7) **Everything else** (Postulates 1, 2, 4; the action itself, Postulate 3; Propositions 3–8; Conjectures 1–2; the numerical, cosmological, Bayesian, and uncertainty sections) is unchanged from v0.6.0 and retains its original epistemic label.

# Contents

# 1 Introduction

## 1.1 Scientific Context and the Scale Incompatibility Problem

Modern physics operates on two foundational pillars that are conceptually and mathematically in conflict:

1. **General Relativity (GR):** A deterministic, background-independent field theory of spacetime geometry, in which dynamics are governed by smooth, continuous pseudo-Riemannian geometry.

2. **Quantum Mechanics (QM) and Quantum Field Theory (QFT):** A probabilistic, operator-based framework in which measurement outcomes are non-deterministic and states undergo unitary time evolution punctuated by wave-function reduction.

Traditional attempts at quantum gravity (String Theory, Loop Quantum Gravity) attempt to quantize spacetime geometry itself. MER Theory approaches the problem from the opposite direction: asking whether the apparent non-determinism and discreteness of quantum phenomena are epistemic artifacts arising from an unexamined scale mismatch between the observer ($O$) and the observed system ($S$).

### 1.2 Epistemic Classification Scheme

To maintain strict mathematical and empirical standards, every statement in this manuscript is classified into one of three epistemic tiers:

- [**Postulate**] (Axiomatic Foundation): an unproven foundational assumption, geometric definition, or variational principle from which dynamics are derived.
- [**Proposition**] (Derived Result): a mathematically rigorous consequence obtained directly from the Postulates via variation, differential geometry, or analytical limits, with the derivation shown explicitly.
- [**Conjecture**] (Physical Hypothesis): a phenomenological interpretation, physical proposal, or experimental correspondence requiring further validation. Conjectures are not derived from the action and should not be read as such.

## 2 Fundamental Postulates

**Postulate 1** (Spacetime Manifold and Kinematic Fields)**.** *Physical reality is defined over a 4-dimensional, connected, smooth pseudo-Riemannian manifold* $(M, g_{\mu\nu})$ *with metric signature* $(-,+,+,+)$*. The kinematic state of the universe is completely described by two smooth field configurations on* $M$*:*

1. *A complex scalar order-parameter field* $\Phi(x^\mu) = \rho(x^\mu)e^{i\theta(x^\mu)} \in \mathbb{C}$*, representing local field amplitude* $\rho$ *and phase* $\theta$*.*

2. *An observer-mismatch vector field* $\epsilon^\mu(x^\mu) \in T_xM$*, representing the spatial, temporal, and energetic scale mismatch between observer and system.*

**Postulate 2** (The Golden Ratio Coupling Eigenvalues)**.** *The non-linear coupling between expansion dynamics and regulatory contraction is governed by the algebraic eigenvalues of the characteristic polynomial* $x^2 - x - 1 = 0$*:*

$$\phi = \frac{1+\sqrt{5}}{2} \approx 1.61803398875\ldots, \qquad \psi = \frac{1-\sqrt{5}}{2} = 1 - \phi = -\frac{1}{\phi} \approx -0.61803398875\ldots \tag{1}$$

*These eigenvalues satisfy the exact algebraic identities:*

$$\phi + \psi = 1, \qquad \phi \cdot \psi = -1, \qquad \phi - \psi = \sqrt{5}, \qquad \phi^2 = \phi + 1, \qquad \psi^2 = \psi + 1 \tag{2}$$

*As shown in Section 5, the scalar field equation of motion depends on $\phi, \psi$ only through the combination $\phi - \psi = \sqrt{5}$. The individual values of $\phi$ and $\psi$ are therefore not separately constrained by that equation; any pair $a, b$ with $a - b = \sqrt{5}$ reproduces it identically. Postulate 2 remains the theory's definition of the coupling, but its specifically golden-ratio character is not, by itself, tested by the scalar dynamics alone.*

# 3 Mathematical Framework

The fundamental mathematical arena is the tangent bundle $TM$ over $(M, g_{\mu\nu})$. Tensor field operations use standard Christoffel connection coefficients:

$$\Gamma^\alpha_{\mu\nu} = \frac{1}{2} g^{\alpha\beta} \left( \partial_\mu g_{\nu\beta} + \partial_\nu g_{\mu\beta} - \partial_\beta g_{\mu\nu} \right) \tag{3}$$

Covariant derivatives on the scalar $\Phi$, the vector $\epsilon^\mu$, and rank-2 tensors follow standard differential geometry:

$$\nabla_\mu \Phi = \partial_\mu \Phi \tag{4}$$

$$\nabla_\mu \epsilon^\nu = \partial_\mu \epsilon^\nu + \Gamma^\nu_{\mu\alpha} \epsilon^\alpha \tag{5}$$

$$\nabla_\mu \epsilon_\nu = \partial_\mu \epsilon_\nu - \Gamma^\alpha_{\mu\nu} \epsilon_\alpha \tag{6}$$

# 4 Action Principle and Lagrangian

**Postulate 3** (The Covariant MER Action Principle)**.** *The complete dynamics of the gravitational metric $g_{\mu\nu}$, scalar order parameter $\Phi$, and scale-mismatch vector $\epsilon^\mu$ are governed by extremization ($\delta S_{MER} = 0$) of the action functional*

$$S_{MER}[g_{\mu\nu}, \Phi, \epsilon_\mu] = \int_M d^4x \sqrt{-g} \left[ \frac{R}{16\pi G} + \mathcal{L}_{\Phi,\epsilon}(g_{\mu\nu}, \Phi, \nabla_\mu \Phi, \epsilon_\mu, \nabla_\mu \epsilon_\nu) \right] \tag{7}$$

*where $R = g^{\mu\nu} R_{\mu\nu}$ is the Ricci scalar, $g = \det(g_{\mu\nu})$, and the matter-scale Lagrangian density is*

$$\mathcal{L}_{\Phi,\epsilon} = -\frac{1}{2} g^{\mu\nu} (\nabla_\mu \Phi^*)(\nabla_\nu \Phi) - \frac{1}{4} F^{(\epsilon)}_{\mu\nu} F^{(\epsilon)\mu\nu} - V(\Phi, \epsilon_\mu) \tag{8}$$

*with the scale field-strength tensor $F^{(\epsilon)}_{\mu\nu} \equiv \nabla_\mu \epsilon_\nu - \nabla_\nu \epsilon_\mu$, and the multi-scale interaction potential*

$$V(\Phi, \epsilon_\mu) = \lambda \left( |\Phi|^2 - v^2 \right)^2 + \frac{1}{2} m_\epsilon^2 \, \epsilon_\mu \epsilon^\mu + \kappa \Big[ \phi \, \epsilon^\mu \nabla_\mu \left( |\Phi|^2 \right) + \psi \, |\Phi|^2 \left( \nabla_\mu \epsilon^\mu \right) \Big] \tag{9}$$

*where $\lambda$ is a dimensionless self-coupling constant, $v$ is the vacuum expectation value, $m_\epsilon$ is the scale-field mass, and $\kappa$ is a dimensionless universal coupling constant.*

## 5 Euler–Lagrange Field Equations

This section has been re-derived in full for the scalar equation. The derivation is shown explicitly (it was previously only asserted) so that the calculation can be checked.

**Proposition 1** (Scalar Order-Parameter Field Equation — Corrected)**.** *Stationarity of $S_{MER}$ with respect to variations $\delta\Phi^*$ yields*

$$\boxed{\Box\Phi - 4\lambda\left(|\Phi|^2 - v^2\right)\Phi + 2\sqrt{5}\,\kappa\,\Phi\,(\nabla_\mu\epsilon^\mu) = 0} \tag{10}$$

*where $\Box \equiv g^{\mu\nu}\nabla_\mu\nabla_\nu$ is the covariant d'Alembertian. Notably,* no term proportional to $\epsilon^\mu\nabla_\mu\Phi$ survives*: it cancels identically against a term generated by the $\phi$-piece of the potential. This differs from the field equation stated in v0.6.0, which retained a spurious $2\phi\,\epsilon^\mu\nabla_\mu\Phi$ term and mis-stated the coefficient of the surviving term.*

*Proof.* Treat $\Phi$ and $\Phi^*$ as independent fields, per the standard trick for complex scalar fields, and vary with respect to $\Phi^*$. Writing out the $\Phi^*$-dependence of $\mathcal{L}_{\Phi,\epsilon}$ term by term:

$$\begin{aligned}
\mathcal{L} \supset & -\tfrac{1}{2}(\nabla_\mu\Phi^*)(\nabla^\mu\Phi) && \text{(kinetic)}\\
& -\lambda(\Phi^*\Phi - v^2)^2 && \text{(symmetry-breaking)}\\
& -\kappa\phi\,\epsilon^\mu\nabla_\mu(\Phi^*\Phi) && \text{($\phi$-term, using $|\Phi|^2 = \Phi^*\Phi$)}\\
& -\kappa\psi\,\Phi^*\Phi\,\nabla_\mu\epsilon^\mu && \text{($\psi$-term)}
\end{aligned}$$

Expanding the $\phi$-term via the product rule, $\nabla_\mu(\Phi^*\Phi) = (\nabla_\mu\Phi^*)\Phi + \Phi^*\nabla_\mu\Phi$, shows that it contributes to *both* $\partial\mathcal{L}/\partial(\nabla_\mu\Phi^*)$ (through the $(\nabla_\mu\Phi^*)\Phi$ piece) and to $\partial\mathcal{L}/\partial\Phi^*$ (through the $\Phi^*\nabla_\mu\Phi$ piece, since $\Phi^*$ itself, undifferentiated, multiplies $\nabla_\mu\Phi$ there). Collecting terms:

$$\frac{\partial\mathcal{L}}{\partial(\nabla_\mu\Phi^*)} = -\tfrac{1}{2}\nabla^\mu\Phi - \kappa\phi\,\epsilon^\mu\Phi \tag{11}$$

$$\frac{\partial\mathcal{L}}{\partial\Phi^*} = -2\lambda(|\Phi|^2 - v^2)\Phi - \kappa\phi\,\epsilon^\mu\nabla_\mu\Phi - \kappa\psi\,\Phi\,\nabla_\mu\epsilon^\mu \tag{12}$$

The Euler–Lagrange equation is $\nabla_\mu[\partial\mathcal{L}/\partial(\nabla_\mu\Phi^*)] - \partial\mathcal{L}/\partial\Phi^* = 0$. Substituting (11)–(12) and expanding $\nabla_\mu(\epsilon^\mu\Phi) = (\nabla_\mu\epsilon^\mu)\Phi + \epsilon^\mu\nabla_\mu\Phi$:

$$-\tfrac{1}{2}\Box\Phi - \kappa\phi\big[(\nabla_\mu\epsilon^\mu)\Phi + \epsilon^\mu\nabla_\mu\Phi\big] + 2\lambda(|\Phi|^2 - v^2)\Phi + \kappa\phi\,\epsilon^\mu\nabla_\mu\Phi + \kappa\psi\,\Phi\,\nabla_\mu\epsilon^\mu = 0$$

The two $\kappa\phi\,\epsilon^\mu\nabla_\mu\Phi$ terms — one from each side — cancel *exactly.* What remains is

$$-\tfrac{1}{2}\Box\Phi + 2\lambda(|\Phi|^2 - v^2)\Phi + \kappa(\psi - \phi)\,\Phi\,\nabla_\mu\epsilon^\mu = 0$$

Multiplying by $-2$ and using $\psi - \phi = -\sqrt{5}$ from (2) gives (10).

As a cross-check, integrate the $\phi$-term of the action by parts before varying at all: for a vector $A^\mu$ and scalar $B$, $\int\sqrt{-g}\,A^\mu\nabla_\mu B = -\int\sqrt{-g}\,(\nabla_\mu A^\mu)B$ up to a boundary term. Applying this to $-\kappa\phi\int\sqrt{-g}\,\epsilon^\mu\nabla_\mu|\Phi|^2$ shows it is equivalent, modulo a boundary term, to $+\kappa\phi\int\sqrt{-g}\,(\nabla_\mu\epsilon^\mu)|\Phi|^2$. Added to the $\psi$-term's contribution, the two $\kappa$-terms in the action are together equivalent to $\kappa(\phi-\psi)\int\sqrt{-g}\,(\nabla_\mu\epsilon^\mu)|\Phi|^2 = \kappa\sqrt{5}\int\sqrt{-g}\,(\nabla_\mu\epsilon^\mu)|\Phi|^2$, a term with *no derivative acting on $\Phi$ at all.* Varying this manifestly $\Phi$-derivative-free form with respect to $\Phi^*$ reproduces (10) directly, with no cancellation needed because there is nothing left to cancel. The two methods agree. □

**Proposition 2** (Scale Vector Field Equation). *Stationarity of $S_{MER}$ with respect to variations $\delta\epsilon_\nu$ yields a non-linear Proca-type vector equation of the schematic form*

$$\nabla_\mu F^{(\epsilon)\mu\nu} - m_\epsilon^2 \epsilon^\nu = \kappa\sqrt{5}\,\nabla^\nu\big(|\Phi|^2\big) \tag{13}$$

***Flag (unverified in this revision):*** *the same mechanism that produces the $\sqrt{5} = \phi - \psi$ factor in Proposition 1 is expected to operate here as well (the $\phi$-term is now the non-derivative piece and the $\psi$-term the derivative piece, so the roles swap relative to the scalar case), and a quick check reproduces a $\sqrt{5}$ coefficient of this form. However, the overall sign was not exhaustively cross-checked against the metric signature and variational conventions used elsewhere in this document, unlike Proposition 1, which was verified by two independent methods. Equation (13) is therefore retained in its v0.6.0 form but should be treated as provisional pending a dedicated re-derivation with the same rigor applied above.*

# 6 Covariant Tensor Formalism

**Proposition 3** (Derivation of the MER Stress-Energy Tensor). *Stationarity of $S_{MER}$ with respect to metric variations $\delta g^{\mu\nu}$ yields $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}^{(MER)}$, where*

$$T_{\mu\nu}^{(MER)} \equiv -\frac{2}{\sqrt{-g}}\frac{\delta(\sqrt{-g}\mathcal{L}_{\Phi,\epsilon})}{\delta g^{\mu\nu}} = T_{\mu\nu}^{(\Phi)} + T_{\mu\nu}^{(\epsilon)} + T_{\mu\nu}^{(int)} \tag{14}$$

*with*

$$T_{\mu\nu}^{(\Phi)} = \tfrac{1}{2}(\nabla_\mu\Phi^*\nabla_\nu\Phi + \nabla_\nu\Phi^*\nabla_\mu\Phi) - g_{\mu\nu}\left[\tfrac{1}{2}g^{\alpha\beta}\nabla_\alpha\Phi^*\nabla_\beta\Phi + \lambda(|\Phi|^2 - v^2)^2\right] \tag{15}$$

$$T_{\mu\nu}^{(\epsilon)} = g^{\alpha\beta}F_{\mu\alpha}^{(\epsilon)}F_{\nu\beta}^{(\epsilon)} - \tfrac{1}{4}g_{\mu\nu}F_{\alpha\beta}^{(\epsilon)}F^{(\epsilon)\alpha\beta} + m_\epsilon^2\epsilon_\mu\epsilon_\nu - \tfrac{1}{2}g_{\mu\nu}m_\epsilon^2\epsilon_\alpha\epsilon^\alpha \tag{16}$$

$$T_{\mu\nu}^{(int)} = -\kappa g_{\mu\nu}\left[\phi\,\epsilon^\alpha\nabla_\alpha|\Phi|^2 + \psi|\Phi|^2\nabla_\alpha\epsilon^\alpha\right] + \kappa\left[\phi\,\epsilon_{(\mu}\nabla_{\nu)}|\Phi|^2 + \psi|\Phi|^2\nabla_{(\mu}\epsilon_{\nu)}\right] \tag{17}$$

*$T_{\mu\nu}^{(int)}$ has not been independently re-derived in this revision (Erratum E5) and should be treated as unverified given the error found in Proposition 1. Full derivations are provided in Appendix A.*

# 7 Observer Scale Operator

**Postulate 4** (Observer Scale Projection Operator). *The observer scale vector field $\epsilon^\mu$ defines an observer scale projection tensor*

$$\Omega_{\mu\nu} = g_{\mu\nu} + \frac{\epsilon_\mu\epsilon_\nu}{\|\epsilon\|^2}, \qquad \|\epsilon\|^2 = g_{\alpha\beta}\epsilon^\alpha\epsilon^\beta \tag{18}$$

*The scalar magnitude $\epsilon \equiv \sqrt{\|\epsilon\|^2}$ quantifies the resolution mismatch ratio $\epsilon = L_S/L_O$.*

- *When $\epsilon \geq 1$ (macro-scale), $\Omega_{\mu\nu}$ collapses to classical Riemannian geometry.*
- *When $\epsilon \ll 1$ (micro-scale), $\Omega_{\mu\nu}$ introduces non-local phase projection metrics, giving rise to epistemic probability distributions $P_\lambda \propto \exp(-\alpha\epsilon^2/\|\Phi\|^2)$.*

## 8 Noether Symmetries and Conserved Currents

**Proposition 4** (Conserved Phase Current). *Global $U(1)$ phase invariance $\Phi \to e^{i\alpha}\Phi$ yields the conserved topological Noether current*

$$J^{\mu}_{(U1)} = \frac{i}{2}\left(\Phi^* \nabla^{\mu} \Phi - \Phi \nabla^{\mu} \Phi^*\right), \qquad \nabla_{\mu} J^{\mu}_{(U1)} = 0 \tag{19}$$

*This derivation was checked independently in this revision (Appendix C) and is unchanged from v0.6.0; it is a standard Noether calculation and is correct as stated.*

**Proposition 5** (Energy-Momentum Conservation). *Local spacetime diffeomorphism invariance guarantees*

$$\nabla_{\mu} T^{(MER)\mu\nu} = 0 \tag{20}$$

## 9 Dimensional Analysis

Evaluating all physical quantities in natural units ($\hbar = c = 1$) confirms structural dimensional consistency:

| Quantity | Symbol | Mass Dimension | SI Units |
|---|---|---|---|
| Metric Tensor | $g_{\mu\nu}$ | 0 | dimensionless |
| Scalar Field | $\Phi$ | 1 | $\mathrm{kg}^{1/2} \cdot \mathrm{m}^{1/2} \cdot \mathrm{s}^{-1}$ |
| Scale Vector Field | $\epsilon^{\mu}$ | 1 | $\mathrm{m}^{-1}$ |
| Field Strength | $F^{(\epsilon)}_{\mu\nu}$ | 2 | $\mathrm{m}^{-2}$ |
| Coupling Constants | $\lambda, \kappa$ | 0 | dimensionless |
| Scale Mass | $m_{\epsilon}$ | 1 | kg ($\mathrm{m}^{-1}$) |
| Energy Density | $T_{\mu\nu}$ | 4 | $\mathrm{J} \cdot \mathrm{m}^{-3}$ |

## 10 Recovery of Classical Mechanics

**Proposition 6** (Macro-Point Particle Geodesic Recovery). *In the limit where $\Phi$ forms a localized spatial wavepacket with envelope mass $M = \int d^3x\, T^{(\Phi)}_{00}$ and the observer operates at macroscopic scales ($\epsilon \to 1$), the center of mass $X^i(t) = \frac{1}{M}\int d^3x\, x^i T_{00}$ obeys Newton's second law in flat space:*

$$M \frac{d^2 X^i}{dt^2} = -\nabla^i V_{ext}(X) + O(\kappa \epsilon^{\mu}) \tag{21}$$

## 11 Recovery of General Relativity

**Proposition 7** (Asymptotic Recovery of Einstein's Field Equations). *When the scale-mismatch field vanishes ($\epsilon^{\mu}(x^{\mu}) \to 0$) and the scalar field relaxes to its vacuum expectation value ($|\Phi| \to v$), $T^{(\epsilon)}_{\mu\nu} \to 0$ and $T^{(int)}_{\mu\nu} \to 0$, reducing the field equations identically to*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\, T^{(baryonic)}_{\mu\nu} \tag{22}$$

## 12 Recovery of Quantum Mechanics

**Proposition 8** (Madelung Hydrodynamic Recovery of Schrödinger's Equation — Corrected)**.** *Expressing $\Phi(x,t) = \sqrt{\rho(x,t)}\exp\left(\frac{iS(x,t)}{\hbar}\right)$ in flat spacetime ($g_{\mu\nu} = \eta_{\mu\nu}$), for non-relativistic velocities and using the corrected scalar equation* (10)*, yields:*

1. ***Continuity Equation (exact, no $O(\kappa)$ correction):***

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\rho \frac{\nabla S}{m}\right) = 0 \tag{23}$$

2. ***Hamilton–Jacobi Equation (carries the $O(\kappa)$ correction):***

$$-\frac{\partial S}{\partial t} = \frac{(\nabla S)^2}{2m} + V_{ext} + Q_{Bohm} - 2\sqrt{5}\,\kappa\hbar\,\frac{\partial \epsilon^0}{\partial t} \tag{24}$$

*where $Q_{Bohm} \equiv -\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}}$. As $\epsilon^0 \to 0$ (and its time derivative with it), recombining $\psi_{QM} = \sqrt{\rho}\,e^{iS/\hbar}$ recovers the Schrödinger equation exactly:*

$$i\hbar\frac{\partial \psi_{QM}}{\partial t} = \left(-\frac{\hbar^2}{2m}\nabla^2 + V_{ext}\right)\psi_{QM} \tag{25}$$

*This differs from the v0.6.0 result, which (incorrectly, per Section 5) placed an $O(\kappa\epsilon^0)$ correction on the continuity equation. The corrected derivation shows the coupling cannot touch probability conservation at all; it can only act like an additional, time-dependent potential term. See Appendix B for the full substitution.*

## 13 Topological Field Solutions

**Conjecture 1** (3D Lemniscate Separatrix and $ER = EPR$)**.** *Localized field concentrations are conjectured to admit non-linear topological separatrix solutions given in polar coordinates by*

$$r^2(\theta) = a^2\cos(2\theta)\cdot\exp\left(\frac{\phi}{\psi}\,\epsilon\right) \tag{26}$$

- ***Central X-Point*** $(0,0)$*: a spatial null throat where opposing vector fluxes cancel.*
- ***Outer O-Points****: dual high-tension field lobes.*
- ***Quantum Entanglement****: entangled particle pairs are conjectured to correspond to the two outer O-lobes of a single unified lemniscate separatrix (EPR), connected through the central X-point throat (ER).*

*This is labeled a Conjecture, not a Proposition, precisely because Eq.* (26) *is stated by analogy rather than derived from the potential $V(\Phi,\epsilon_\mu)$ of Eq.* (9) *or from the field equations of Section 5. No derivation connecting this ansatz to the action is currently available.*

## 14 Numerical Simulation Framework & Benchmarks

### 14.1 Computational Architecture

- **Spatial Discretization:** 3D uniform spatial grid of size $N_x \times N_y \times N_z = 512^3$ with periodic boundary conditions.
- **Integration Method:** Pseudo-spectral split-operator method with Fast Fourier Transforms (FFTW3).
- **Time-Stepping Scheme:** 4th-order Runge–Kutta–Fehlberg (RKF45) with adaptive CFL condition $\Delta t \leq \eta \dfrac{\Delta x}{c\sqrt{1 + \|\epsilon^\mu\|_{max}}}$.

### 14.2 Code Validation Suite (Proposed Verification Benchmarks)

Prior to analyzing physical scenarios, the solver must pass three standardized mathematical verification tests:

1. **Linear Limit Conservation:** in the limit $\lambda \to 0$, $\kappa \to 0$, energy $P^0$ must be conserved to machine precision ($\Delta P^0/P^0 < 10^{-14}$).
2. **Solitonic Convergence:** spatial grid refinement tests must confirm $L_2$-error convergence scaling as $O(\Delta x^P)$ for spectral expansion order $P$.
3. **Gauge Conservation:** the vector constraint $\nabla_\mu \epsilon^\mu = 0$ must hold to within $\delta < 10^{-10}$ across all integration time steps.

Because the corrected scalar equation (Eq. (10)) couples to $\nabla_\mu \epsilon^\mu$ rather than to $\epsilon^\mu \nabla_\mu \Phi$, any solver implementing Proposition 1 should be updated to match; a solver built against the v0.6.0 field equation would not be integrating the equation this document now derives.

## 15 Cosmological Applications (Proposed Computational Studies)

**Conjecture 2** (Bullet Cluster Lensing Peak Separation)**.** *High-velocity galactic cluster collisions (e.g. 1E 0657-558) can be modeled using the MER non-linear field solver to test whether intergalactic hydrogen gas stalls at the central X-point due to electromagnetic pressure, while non-linear field stress* $T^{(int)}_{\mu\nu}$ *carries spacetime curvature forward into two outer O-lobes, producing gravitational lensing offsets without cold dark matter particles.*

## 16 Experimental Predictions & Falsification Criteria

This section has been reframed. In v0.6.0 the table below was presented as though it specified numerical falsification thresholds. As stated, none of the four rows carries an explicit proportionality constant, and Section 17.2 places only a uniform prior $\kappa \in [0, 1]$ on the coupling — so no observation could yet be said to rule the theory in or out at a stated confidence level. The table is retained as a statement of the *functional form* MER predicts for each signal; turning any row into an actual falsification threshold requires (i) deriving the proportionality constant from the field equations (e.g. linearizing $G_{\mu\nu} = 8\pi G\, T^{(MER)}_{\mu\nu}$ around a gravitational-wave background to get an explicit dispersion relation), and (ii) an independent constraint on $\kappa$ and $\epsilon^\mu$ for the physical system in question. Neither step has been carried out here.

| Phenomenon | Standard Physics | MER Predicted Form | Platform | Status |
|---|---|---|---|---|
| GW Phase Lag | Pure GR inspiral ($\delta\Psi_{GW} = 0$) | $\delta\Psi_{GW} \propto f^{-5/3}(\kappa\epsilon^\mu)^2$ | LIGO O5 / LISA | Functional form only; proportionality constant not yet derived. |
| Decoherence Floor | Rate scales $t_d \propto T^{-1}$ | $t_d^{min} = \dfrac{\hbar}{E_S\,\epsilon^0\phi}$ | Atom Interferometry | Functional form only; requires independent $\epsilon^0$ estimate. |
| CMB B-Mode | Primordial GWs | Anisotropic peak shift at $\ell \approx 3560$ | CMB-S4 | Peak location stated; predicted amplitude not derived. |
| Galactic Lensing | Lensing tracks WIMPs | Offset $\Delta x \propto \kappa\sqrt{5}\,v^2/m_\epsilon$ | JWST / Euclid | Functional form only; proportionality constant not yet derived. |

## 17 Proposed Bayesian Parameter Estimation Pipeline

To estimate the universal coupling parameters $\Theta = \{\kappa, m_\epsilon, \lambda, v\}$ once observational data pipelines are integrated, we formulate a Bayesian MCMC sampling protocol.

### 17.1 Bayesian Likelihood Formulation

Given an observational dataset $D = \{y_i, \sigma_i\}_{i=1}^N$, the log-likelihood is

$$\ln p(D \mid \Theta) = -\frac{1}{2}\sum_{i=1}^{N}\left[\frac{(y_i - y_{MER}(x_i;\Theta))^2}{\sigma_i^2 + \sigma_{sys}^2} + \ln 2\pi(\sigma_i^2 + \sigma_{sys}^2)\right] \tag{27}$$

where $y_{MER}(x_i;\Theta)$ is the numerical evaluation of the MER field equations at observation point $x_i$.

### 17.2 Proposed MCMC Protocol & Mock Sampling Strategy

- **Sampler:** Ensemble slice sampling / Metropolis–Hastings MCMC ($10^6$ iterations, $10^4$ burn-in steps).
- **Prior Distributions:** Uninformative uniform prior on $\kappa \in [0,1]$, log-uniform on $m_\epsilon \in [10^{-25}, 10^{-15}]\,\mathrm{eV}$.
- **Objective:** map parameter degeneracy contours (e.g. correlation $\rho_{\kappa,m_\epsilon}$) to quantify trade-offs between scale vector mass and coupling strength.

As noted in Section 16, this wide a prior on $\kappa$ is itself evidence that the theory has not yet produced a sharp numerical prediction for any of the falsification-table rows; that is precisely what this MCMC pipeline is intended to eventually narrow down, given data.

## 18 Uncertainty Analysis Framework

Uncertainty propagation from observational errors $\sigma_i$ to parameter posteriors is computed via the covariance matrix

$$\Sigma_{ij} = \left\langle (\Theta_i - \bar{\Theta}_i)(\Theta_j - \bar{\Theta}_j) \right\rangle \tag{28}$$

This statistical pipeline is intended to ensure that future empirical data produce rigorous 68% ($1\sigma$) and 95% ($2\sigma$) confidence intervals for all MER coupling constants.

## 19 Discussion

The MER framework addresses the long-standing conceptual rift between quantum mechanics and general relativity by shifting the focus from ontology to epistemology. Rather than quantizing space or modifying gravity with unobserved dark matter particles, MER proposes that both regimes emerge as scale-dependent asymptotic projections of a single covariant field action $S_{MER}$.

Two points from this revision are worth stating plainly. First, the corrected scalar field equation (Proposition 1) is, if anything, a more defensible result than the one it replaces: rather than perturbing probability conservation — which would have been a serious conceptual problem for any theory purporting to recover standard quantum mechanics — the scale-mismatch coupling only ever modifies the phase/energy sector, exactly the sector where a new potential-like term is comparatively unremarkable. Second, the fact that only $\phi - \psi = \sqrt{5}$ (and not $\phi$ and $\psi$ individually) enters the scalar dynamics means the golden-ratio framing of Postulate 2, however aesthetically motivated, is not by itself supported by the scalar sector; it is a stronger claim than the mathematics here currently requires. Whether the vector sector (Proposition 2) or the stress tensor (Appendix A) depend on $\phi, \psi$ individually rather than only through $\sqrt{5}$ is an open question this revision does not settle.

## 20 Conclusion

MER Theory v0.7.0 provides a mathematically corrected, internally consistent scalar sector, together with an explicit account of what remains provisional. By re-deriving the scalar field equation and the Madelung limit from the stated action, and by being explicit about which parts of the theory (the vector equation, the stress tensor, the falsification thresholds) have not yet been independently verified, this revision aims to leave a clean record of what is established, what is provisional, and what remains conjectural, so that future versions can build on a checked foundation rather than propagate an unverified one.

## A Unabridged Metric Variation and Field Equation Proofs

Unchanged from v0.6.0 (Erratum E5); not independently re-derived in this revision.

We perform the explicit variation of the matter action $S_{matter} = \int d^4x \sqrt{-g}\, \mathcal{L}_{\Phi,\epsilon}$ with respect to the inverse metric $g^{\mu\nu}$. The defining expression for the stress-energy tensor is

$$T^{(MER)}_{\mu\nu} \equiv -\frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g}\mathcal{L}_{\Phi,\epsilon})}{\delta g^{\mu\nu}} = -2\frac{\partial \mathcal{L}_{\Phi,\epsilon}}{\partial g^{\mu\nu}} + g_{\mu\nu}\mathcal{L}_{\Phi,\epsilon} \tag{29}$$

### A.1 Scalar Field Component ($T^{(\Phi)}_{\mu\nu}$)

The scalar kinetic Lagrangian is $\mathcal{L}^{(\Phi)}_{kin} = -\frac{1}{2}g^{\alpha\beta}\nabla_\alpha\Phi^*\nabla_\beta\Phi$. Differentiating with respect to $g^{\mu\nu}$:

$$\frac{\partial \mathcal{L}^{(\Phi)}_{kin}}{\partial g^{\mu\nu}} = -\tfrac{1}{2}\nabla_\mu\Phi^*\nabla_\nu\Phi \tag{30}$$

Multiplying by $-2$ and adding $g_{\mu\nu}\mathcal{L}_\Phi$ gives Eq. (15).

### A.2 Vector Field Component ($T^{(\epsilon)}_{\mu\nu}$)

The scale-vector kinetic term is $\mathcal{L}^{(\epsilon)}_{kin} = -\frac{1}{4}g^{\alpha\gamma}g^{\beta\delta}F_{\alpha\beta}F_{\gamma\delta}$. Differentiating with respect to $g^{\mu\nu}$:

$$\frac{\partial \mathcal{L}^{(\epsilon)}_{kin}}{\partial g^{\mu\nu}} = -\tfrac{1}{2}g^{\beta\delta}F_{\mu\beta}F_{\nu\delta} = -\tfrac{1}{2}F_{\mu\alpha}F_\nu{}^\alpha \tag{31}$$

Multiplying by $-2$ yields the standard gauge-field stress tensor $F_{\mu\alpha}F_\nu{}^\alpha - \frac{1}{4}g_{\mu\nu}F^2$. Including the mass potential term $\frac{1}{2}m_\epsilon^2 g^{\alpha\beta}\epsilon_\alpha\epsilon_\beta$ gives Eq. (16).

### A.3 Interaction Component ($T^{(int)}_{\mu\nu}$)

The interaction potential contains the derivative terms $\kappa[\phi\,\epsilon^\alpha\nabla_\alpha|\Phi|^2 + \psi|\Phi|^2\nabla_\alpha\epsilon^\alpha]$. Applying symmetric index projection $\delta g^{\alpha\beta}\epsilon_\alpha = \epsilon^{(\mu}\delta g^{\nu)}{}_\beta$ gives Eq. (17). Combining all three components yields the tensor stated in Eq. (14). ■

## B Step-by-Step Proof of the Madelung Quantum Limit — Corrected

This appendix has been re-derived to match the corrected scalar equation, Eq. (10). The kinetic-term algebra (gradient and time-derivative identities for the polar ansatz) is unchanged from v0.6.0, since that part of the calculation does not involve $\kappa, \phi, \psi$; only the treatment of the coupling term differs.

In flat Minkowski space ($g_{\mu\nu} = \eta_{\mu\nu}$), the corrected scalar equation reads

$$\left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2\right)\Phi - 4\lambda(|\Phi|^2 - v^2)\Phi + 2\sqrt{5}\,\kappa\,\Phi\,(\nabla_\mu\epsilon^\mu) = 0 \tag{32}$$

Substitute the polar representation $\Phi(x,t) = \sqrt{\rho(x,t)}\exp\!\left(\frac{iS(x,t)}{\hbar}\right)$, and, following the same non-relativistic reduction used in v0.6.0 (the $O(1/c^2)$ term is subleading and dropped in the same way), approximate $\nabla_\mu\epsilon^\mu \approx \partial\epsilon^0/\partial t$ (retaining only the temporal mismatch component, as in the original treatment).

### B.1 Why the Coupling Term Cannot Enter the Continuity Equation

The key structural difference from v0.6.0 is that the surviving coupling term, $2\sqrt{5}\kappa\Phi(\partial_t\epsilon^0)$, is *directly proportional to* $\Phi$ *itself* with a real coefficient — it contains no derivative of $\Phi$. The continuity (imaginary-part) equation is obtained from the combination $\Phi^* \times$ (field equation) $- \Phi \times$ (field equation)$^*$. For any term of the form $R(x,t)\,\Phi$ with $R$ real,

$$\Phi^*\big[R\Phi\big] - \Phi\big[R\Phi^*\big] = R\,(\Phi^*\Phi - \Phi\Phi^*) = 0 \tag{33}$$

identically. Hence the coupling term contributes *nothing* to the imaginary part, and the continuity equation is recovered exactly:

$$\frac{\partial\rho}{\partial t} + \nabla\cdot\left(\rho\,\frac{\nabla S}{m}\right) = 0 \tag{34}$$

with no $O(\kappa)$ correction of any kind. This is the opposite of the v0.6.0 result, which placed a spurious $O(\kappa\epsilon^0)$ term here; that term depended on the now-cancelled $\epsilon^\mu\nabla_\mu\Phi$ piece, which carried an imaginary component (through $\nabla_\mu\Phi \ni i\nabla_\mu S\,\Phi/\hbar$) that the corrected, purely-real coupling term does not have.

### B.2 Real Part (Hamilton–Jacobi / Phase Terms)

The real-part (phase) equation is obtained from $\Phi^*\times$(field equation)$+\Phi\times$(field equation)$^*$, divided by $2\rho$. The kinetic and self-coupling contributions reproduce the standard Bohm potential terms as in v0.6.0; the new coupling term contributes an additional real, additive piece proportional to $\sqrt{5}\kappa\hbar\,\partial_t\epsilon^0$. Collecting all terms:

$$-\frac{\partial S}{\partial t} = \frac{(\nabla S)^2}{2m} + V_{ext} - \frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}} - 2\sqrt{5}\,\kappa\hbar\,\frac{\partial\epsilon^0}{\partial t} \tag{35}$$

Recognizing $Q_{Bohm} \equiv -\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}}$ and setting $\psi_{QM} = \sqrt{\rho}\,e^{iS/\hbar}$, this recombines into

$$i\hbar\frac{\partial\psi_{QM}}{\partial t} = \left(-\frac{\hbar^2}{2m}\nabla^2 + V_{ext} - 2\sqrt{5}\,\kappa\hbar\,\frac{\partial\epsilon^0}{\partial t}\right)\psi_{QM} \tag{36}$$

As $\epsilon^0 \to 0$ (and hence $\partial_t\epsilon^0 \to 0$), this recovers the ordinary Schrödinger equation exactly, completing the corrected proof of Proposition 8. ■

## C Complete Noether Current Derivations

Unchanged from v0.6.0; this derivation was checked independently in this revision and found correct.

### C.1 $U(1)$ Phase Invariance Proof

Let $\Phi \to \Phi' = e^{i\alpha}\Phi$. The Lagrangian $\mathcal{L}_{\Phi,\epsilon}$ depends on $\Phi$ only through $|\Phi|^2 = \Phi^*\Phi$ and the kinetic term $(\nabla_\mu\Phi^*)(\nabla^\mu\Phi)$.

1. $\delta|\Phi|^2 = (e^{-i\alpha}\Phi^*)(e^{i\alpha}\Phi) - \Phi^*\Phi = 0$.
2. $\delta(\nabla_\mu\Phi^*\nabla^\mu\Phi) = (\nabla_\mu(e^{-i\alpha}\Phi^*))(\nabla^\mu(e^{i\alpha}\Phi)) - \nabla_\mu\Phi^*\nabla^\mu\Phi = 0$.

By Noether's first theorem, the conserved current associated with the continuous symmetry parameter $\alpha$ is

$$J^\mu = \frac{\partial\mathcal{L}}{\partial(\nabla_\mu\Phi)}\frac{\delta\Phi}{\delta\alpha} + \frac{\partial\mathcal{L}}{\partial(\nabla_\mu\Phi^*)}\frac{\delta\Phi^*}{\delta\alpha} \tag{37}$$

Evaluating the partial derivatives,

$$\frac{\partial \mathcal{L}}{\partial(\nabla_\mu \Phi)} = -\tfrac{1}{2}\nabla^\mu \Phi^*, \quad \frac{\delta \Phi}{\delta \alpha} = i\Phi, \qquad \frac{\partial \mathcal{L}}{\partial(\nabla_\mu \Phi^*)} = -\tfrac{1}{2}\nabla^\mu \Phi, \quad \frac{\delta \Phi^*}{\delta \alpha} = -i\Phi^* \tag{38}$$

Substituting back,

$$J^\mu_{(U1)} = -\tfrac{1}{2}\nabla^\mu \Phi^*(i\Phi) - \tfrac{1}{2}\nabla^\mu \Phi(-i\Phi^*) = \frac{i}{2}\left(\Phi^* \nabla^\mu \Phi - \Phi \nabla^\mu \Phi^*\right) \tag{39}$$

Taking the covariant divergence, $\nabla_\mu J^\mu_{(U1)} = 0$, completing the proof of Proposition 4. ■